

Twin Decision Theory

Group-theoretic discrimination for twin operation spaces

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Abstract

We give a rigorous, unified framework casting the selection of a twin operation as a *decision problem*, and we promote the structural invariants of group theory — conjugacy classes, centralizers, the centre, and the *class equation* — into instruments of discrimination. We first correct the classification statement of the foundations volume: the operation space Op is canonically the homogeneous space G/L (a coset space), and it carries a group structure if and only if the structural kernel L is *normal*. We exhibit a concrete counterexample (S_4 acting on $+ \bmod 4$, where L has order 2 and is not normal). We then prove the reduction principles: a cost that is a class function is constant on conjugacy classes, so the class equation *enumerates the decision cells*; in general the cells are the orbits of the normalizer $N_G(L)$, counted by Burnside. Finally we record the compact-Lie instance, where the class equation becomes the Weyl integration formula (worked out for $SU(2)$). Every statement is matched to a computational test in `arithmetics/symbolic/twin/decision.py`.

1 The twin decision problem

Definition 1 (Twin decision problem). A *twin decision problem* is a tuple $\mathcal{D} = (E, \odot_e, G, \rho, J)$ where $(E, \odot_e)$ is a magma, $\rho : G \rightarrow \text{Sym}(E)$ is a group action (written $g \cdot x$), and for each $g \in G$ the *twin operation* is

$$x \odot_{eg} y := g^{-1} \cdot ((g \cdot x) \odot_e (g \cdot y)).$$

Let $\pi : G \rightarrow \text{Op}(E, \odot_e; G)$, $g \mapsto \odot_{eg}$, be the *transport map* onto the set Op of distinct twin operations, and

$$L := \{g \in G : \odot_{eg} = \odot_e\}$$

the *structural kernel*. Finally $J : \text{Op} \rightarrow \mathbb{R}_{\geq 0}$ is a *cost*. To *solve* $\mathcal{D}$ is to compute $\odot^\star = \arg \min_{\text{Op}} J$.

All earlier applications — batched arithmetic (speed), dynamic range, function approximation, optimal-base selection, twin differential calculus — are instances of $\mathcal{D}$ with different G and J . This is the single formulation sought.

2 The kernel and the corrected classification

Proposition 2 (Kernel). *L is a subgroup of G , equal to the automorphism group of $(E, \odot_e)$ inside $\rho(G)$.*

Proposition 3 (Fibres are right cosets). *For $g, h \in G$,*

$$\odot_{eg} = \odot_{eh} \iff hg^{-1} \in L \iff Lg = Lh.$$

Proof. $\odot_{eg} = \odot_{eh}$ means $g^{-1} \cdot ((g \cdot x) \odot_e (g \cdot y)) = h^{-1} \cdot ((h \cdot x) \odot_e (h \cdot y))$ for all x, y . Apply $h \cdot (-)$ and substitute $u = g \cdot x$, $v = g \cdot y$ (a bijection of E); with $m := hg^{-1}$ this reads $m \cdot (u \odot_e v) = (m \cdot u) \odot_e (m \cdot v)$ for all u, v , i.e. $m \in L$. Hence $hg^{-1} \in L$, equivalently $Lg = Lh$. $\square$

Corollary 4 (Classification). *The transport map induces a bijection $L \backslash G \xrightarrow{\sim} \text{Op}$, $Lg \mapsto \odot_{eg}$. In particular $|\text{Op}| = [G : L]$.*

Theorem 5 (Group structure $\iff$ normality). *The rule $\odot_{eg} * \odot_{eh} := \odot_{e,gh}$ is a well-defined binary operation on Op (making π a homomorphism and $\text{Op} \cong G/L$ a group) if and only if $L \trianglelefteq G$.*

Proof. Replace g, h by ag, bh with $a, b \in L$. Then $L(ag)(bh) = Lgbh$, which equals Lgh for all $b \in L$ iff $gbg^{-1} \in L$ for all $b \in L$ and all g , i.e. iff $g^{-1}Lg = L$ for all g — precisely $L \trianglelefteq G$. $\square$

Remark 6 (Correction to the foundations volume). The foundations volume asserts that π is always a group homomorphism and hence $L \trianglelefteq G$. This is *false in general*. For $G = S_4$ acting on $E = \{0, 1, 2, 3\}$ with base operation $+$ mod 4, the kernel L has order 2 and is *not* normal (a transposition subgroup; S_4 has no normal subgroup of order 2). Thus Op is the homogeneous space G/L with $|\text{Op}| = 12$, but is *not* a group. The correct universal statement is Corollary 4; the group structure is the special case of Theorem 5.

3 Reduction principles for decision

Definition 7 (Class function). For a finite group H , a cost $J : H \rightarrow \mathbb{R}$ is a *class function* if $J(w x w^{-1}) = J(x)$ for all $w, x \in H$ (invariance under conjugation).

Proposition 8 (Class-function reduction). *If $L \trianglelefteq G$ (so $\text{Op} \cong G/L$ is a group) and J is a class function on Op , then J is constant on each conjugacy class. Consequently $\arg \min J$ is a union of conjugacy classes, and it suffices to evaluate J on one representative per class. The number of evaluations is the number of classes, and the class equation*

$$|\text{Op}| = |Z(\text{Op})| + \sum_i [\text{Op} : C_{\text{Op}}(x_i)]$$

enumerates the decision cells together with their multiplicities.

In the general (non-normal) case the relevant symmetry is the normalizer.

Theorem 9 (Normalizer orbit decomposition). *The conjugation $\odot_{eg} \mapsto \odot_{e, wgw^{-1}}$ is well-defined on Op exactly for $w \in N_G(L)$. The induced action of $N_G(L)$ partitions Op into decision cells (orbits), counted by Burnside:*

$$\# \text{cells} = \frac{1}{|N_G(L)|} \sum_{w \in N_G(L)} |\text{Fix}_{\text{Op}}(w)|.$$

When $L \trianglelefteq G$ one has $N_G(L) = G$ and the cells are exactly the conjugacy classes of $\text{Op} = G/L$, recovering Proposition 8.

Proof. For $w \in N_G(L)$ and $a \in L$, $w(ag)w^{-1} = (waw^{-1})(wgw^{-1})$ with $waw^{-1} \in L$, so $Lwgw^{-1}$ depends only on Lg ; the map is well-defined on Op by Corollary 4. For $w \notin N_G(L)$ there is $a \in L$ with $waw^{-1} \notin L$, and the image depends on the coset representative. Orbit counting is Burnside's lemma; the normal case gives $N_G(L) = G$ acting on G/L by conjugation, whose orbits are the conjugacy classes. $\square$

Remark 10. The quotient $N_G(L)/L$ plays the role of a *Weyl group* of the configuration: it is the largest symmetry acting on Op through the kernel.

4 The class equation as a decision stratification

Read as a spectrum of symmetry, the class equation stratifies the decision space:

- the **centre** $Z(\text{Op})$ consists of operations fixed by every inner relabeling — the frame-canonical, robust choices;
- a **large class** ($[\text{Op} : C(x)]$ large, small centralizer) is a generic, low-symmetry operation;
- a **small class** (large centralizer) is a highly symmetric, structured operation.

Remark 11 (Scope of structural reduction). By the equivariance principle every twin operation is isomorphic to $\odot_e$; hence a cost invariant under *all* relabelings is constant on Op and discriminates nothing. The class equation discriminates exactly the costs that are invariant under the *inner* symmetries (conjugation), such as the order complexity $\kappa_{\text{ord}}(\odot_{eg}) = \text{ord}(gL)$. Frame-dependent costs (conditioning in fixed coordinates, operation count) are handled by their own symmetry group via Theorem 9.

5 Compact Lie groups: the Weyl integration formula

For a compact connected Lie group G with maximal torus T and Weyl group W , the finite class equation becomes the *Weyl integration formula*: for a class function J ,

$$\int_G J d\mu_{\text{Haar}} = \frac{1}{|W|} \int_T J(t) |\Delta(t)|^2 dt,$$

the continuous enumeration of conjugacy classes.

Proposition 12 ($SU(2)$ instance). *Conjugacy classes of $SU(2)$ are parametrized by the angle $\theta \in [0, \pi]$ (eigenvalues $e^{\pm i\theta}$), and the push-forward of the Haar measure is*

$$d\mu(\theta) = \frac{2}{\pi} \sin^2 \theta d\theta, \quad \int_0^\pi d\mu = 1.$$

A class cost $J(\theta)$ is decided by its Weyl average $\int_0^\pi J(\theta) d\mu(\theta)$ and its optimum over $\theta \in [0, \pi]$.

This is the exact analogue of $\frac{1}{|G|} \sum_i |\text{class}_i| J(x_i)$: the class equation, made continuous.

Proposition 13 (Qudits: the $U(n)/SU(n)$ formula). *For $U(n)$ the maximal torus is the diagonal subgroup, the Weyl group is S_n , and the eigen-phase joint density is the squared Vandermonde*

$$d\mu(\theta_1, \dots, \theta_n) = \frac{1}{n! (2\pi)^n} \prod_{j < k} |e^{i\theta_j} - e^{i\theta_k}|^2 d\theta,$$

the CUE eigenvalue density. The irreducible characters are the Schur polynomials $\chi_\lambda = s_\lambda(e^{i\theta})$, and they are orthonormal, $\langle \chi_\lambda, \chi_\mu \rangle = \delta_{\lambda\mu}$ — the relation that pins down the measure. A class cost on a qudit gate (d -level, under conjugation by $SU(d)$) is a symmetric function of the eigen-phases, decided against $d\mu$.

Remark 14 (Universality persists). A generic bilinear gate on $\mathbb{C}^d$ has a trivial structural kernel under $SU(d)$, so $\text{Op} \cong SU(d)$ for every d tested — the $SU(2)$ universality instance extends to qudits.

6 Primary decomposition of the decision space (abelian case)

When G is finite abelian the kernel L is automatically normal, so $\text{Op} = G/L$ is an abelian group and decomposes along its primary (Sylow- p) components. Write $|G| = \prod_i p_i^{a_i}$ and let G_p be the (unique) Sylow p -subgroup.

Theorem 15 (Primary decomposition). *Let G be finite abelian acting on $(E, \odot_e)$ with kernel L . Then $G \cong \prod_p G_p$, every subgroup decomposes as $L = \prod_p L_p$ with $L_p = L \cap G_p$ (so the compatibility hypothesis is automatic), and*

$$\text{Op} = G/L \cong \prod_p G_p/L_p = \prod_p \text{Op}_p.$$

Proof. $G \cong \prod_p G_p$ is the primary decomposition of a finite abelian group; the projection onto G_p is $g \mapsto g^{e_p}$ where $e_p \equiv 1 \pmod{p^{a_p}}$ and $e_p \equiv 0 \pmod{|G|/p^{a_p}}$, whence $\prod_p g^{e_p} = g$. A subgroup of $\prod_p G_p$ that is a product of p -groups is the product of its intersections with the factors, giving $L = \prod_p L_p$. Quotienting factorwise yields $G/L \cong \prod_p G_p/L_p$. $\square$

Corollary 16 (Unique factorisation of operations). *Every twin operation factors uniquely as $\odot_{eg} \leftrightarrow (g_p L_p)_p \in \prod_p \text{Op}_p$, and $|\text{Op}| = \prod_p |\text{Op}_p|$.*

Proposition 17 (Separable decision). *If the cost is separable, $J(\odot_{eg}) = \sum_p J_p(g_p L_p)$, then*

$$\min_{\text{Op}} J = \sum_p \min_{\text{Op}_p} J_p,$$

so the optimum is found componentwise. The search collapses from $\prod_p |\text{Op}_p|$ to $\sum_p |\text{Op}_p|$ candidates — the Chinese Remainder / Fourier factorisation of the decision problem.

Theorem 18 (Order-complexity partition). *Let $n = |\text{Op}|$ and $C_d = \{\odot \in \text{Op} : \kappa_{\text{ord}}(\odot) = d\}$. Then*

$$\text{Op} = \bigsqcup_{d|n} C_d, \quad |C_d| = \varphi(d) m_d,$$

where m_d is the number of cyclic subgroups of Op of order d , and $\sum_{d|n} |C_d| = n$.

Proof. Every $\odot \in \text{Op}$ has an order $\kappa_{\text{ord}}(\odot) = \text{ord}(gL)$ dividing $n = |\text{Op}|$ (Lagrange), so $\text{Op} = \bigsqcup_{d|n} C_d$ and $\sum_{d|n} |C_d| = n$. The elements of order d are exactly the generators of the cyclic subgroups of order d : each such subgroup has $\varphi(d)$ generators, and two order- d elements generate the same subgroup iff each is a power of the other. Summing over the m_d cyclic subgroups of order d gives $|C_d| = \varphi(d) m_d$. $\square$

7 Arithmetic-task costs (making the cost operational)

The costs above were abstract. We now record *measurable* costs tied to genuine computation, taking $G = \langle \varphi \rangle$ a monogenic group so the group element *is* the computational domain. For a bijection φ the twin reduction of data (x_i) is $\bigoplus_{\varphi} x_i = \varphi^{-1}(\sum_i \varphi(x_i))$, with transformed total $T = \sum_i \varphi(x_i)$. For $\varphi = \log$ this is the product, and T is the log-likelihood.

Two well-posed decisions arise, each with a numeric cost J that the tool *measures* rather than postulates:

- **Domain of computation** (fixed target T): evaluate T in the transformed domain $(\sum \varphi(x_i), \text{stable})$ or the original domain $(\varphi(\bigoplus_{\varphi} x_i), \text{prone to overflow/underflow})$. Cost: relative error against a high-precision reference. For $\varphi = \log$ on wide-range data the original domain underflows ($\prod x_i \rightarrow 0, \log 0 = -\infty$) while the transformed domain is exact to rounding.

- **Optimal base** (representation): choose φ so the data is twin-linear; cost is the back-transformed fit RMSE. A power law selects log–log, an exponential selects semi-log, a line selects the identity.

Remark 19 (Honest status). These are *not theorems* but the classical log-domain and linearisation facts of numerical computing. The framework’s contribution is to organise them as a *base-selection decision over $\langle\varphi\rangle$* , made decidable by measurement; there is an honest speed/stability trade-off (the transformed domain controls dynamic range at the price of more transcendental calls).

8 Computational realization

The module `arithmetics/symbolic/twin/decision.py` implements these statements; the test suite `arithmetics/tests/test_twin_decision.py` verifies them. The correspondence is:

Statement	Code	Test
Class equation $ G = Z + \sum[G : C]$	<code>class_equation</code>	<code>TestClassEquation</code>
Class-function reduction (Prop. 8)	<code>is_class_function, decision_report</code>	<code>TestClassFunction</code>
Classification $ \text{Op} = [G : L]$ (Cor. 4)	<code>twin_decision</code>	<code>test_invariant</code>
Group $\iff$ normal (Thm. 5, Rem. 6)	<code>is_normal, quotient</code>	<code>TestNormal</code>
$S_4/+ \bmod 4$ counterexample	<code>twin_decision</code>	<code>test_s4</code>
Normalizer orbits (Thm. 9)	<code>normalizer, operation_orbits</code>	<code>test_orbits</code>
Weyl integration on $SU(2)$ (Prop. 12)	<code>su2_class_average, su2_class_decision</code>	<code>TestWeyl</code>
$U(n)/SU(n)$ Weyl + characters (Prop. 13)	<code>weyl_average, verify_character_orthonormality</code>	<code>TestWeyl</code>
Qudit universality (Rem. after Prop. 13)	<code>qudit_universality</code>	<code>TestQudit</code>
Primary decomposition (Thm. 15)	<code>verify_primary_decomposition</code>	<code>TestOperation</code>
Unique factorisation (Cor. 16)	<code>operation_factorization</code>	<code>test_unique</code>
Separable decision (Prop. 17)	<code>primary_decision</code>	<code>TestSeparable</code>
Complexity partition (Thm. 18)	<code>verify_complexity_partition</code>	<code>TestComplexity</code>
Domain of computation (§ arith.)	<code>total_error, recommend_domain</code>	<code>TestTransformation</code>
Optimal base (§ arith.)	<code>optimal_base</code>	<code>TestOptimal</code>

9 Outlook

The framework generalizes verbatim to abelian groups and direct products (where L is automatically normal and the class equation is the full enumeration), to richer non-abelian groups (dihedral, symmetric, quaternion — where the class equation discriminates strongly), and to compact Lie groups via the Weyl formula. The primary decomposition for abelian G (Section 15ff.) is realised; the remaining steps are: (i) cost functionals tied to genuine arithmetic tasks (conditioning, operation count); and (ii) the $SU(n)$ Weyl formula for the qudit instance.